HOME PLOTS APPS FIGURE FORMAT TOOLS

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New Script | New Live Script | New | Open | Go to File | Find Files | Import Data | Save Workspace | Clean Data | New Variable | Open Variable | Clear Workspace | Favorites | Clear Commands | Simulink | Layout | Set Path | Add-Ons | Settings | Connections | Parallel | Help | Request Support | Feedback | Learn MATLAB | Copilot Chat

FILE VARIABLE CODE SIMULINK ENVIRONMENT RESOURCES COPILOT

/ > MATLAB Drive

Files

Name

- untitled8.m
- untitled7.m
- untitled6.m
- untitled5.m
- untitled4.m
- untitled3.m
- untitled2.m
- untitled01.m
- untitled.m mfw.m
- untitled.m

Workspace

Name	Value	Size	Class
alloc	[10,9,9,9,9,9]	1×6	double
bandwidth	10000000	1×1	double
base	9	1×1	double
d	2	1×1	double
f	[900,3500,28...	1×3	double
PL	[97.5455,109....	1×3	double
rb_size	180000	1×1	double
total_RB	55	1×1	double
users	[1,2,3,4,5,6]	1×6	double
utilization	[18.1818,16.3...	1×6	double

Editor

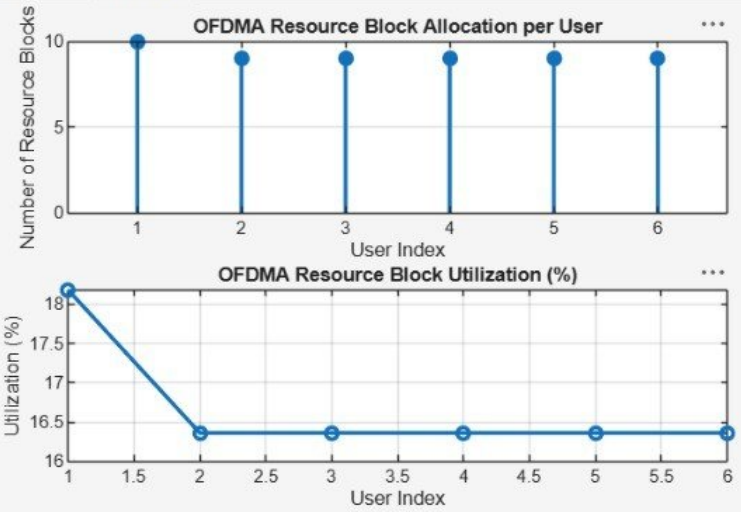
untitled01.m | untitled5.m | untitled2 | untitled6.m | untitled7.m | untitled8.m

/MATLAB Drive/untitled8.m

```
1   % System parameters
2   bandwidth = 10e6; % 10 MHz
3   rb_size = 180e3; % 180 kHz per RB
4   % Total Resource Blocks
5   total_RB = floor(bandwidth / rb_size);
6   % Users
7   users = 1:6;
8   % RB allocation (simple equal + remainder distribution)
9   base = floor(total_RB / length(users));
10
11  alloc = base * ones(1,length(users));
12  alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,l
13  utilization = (alloc / total_RB) * 100;
14  disp('Total Resource Blocks:')
15  disp(total_RB)
16  disp('RB Allocation per User:')
17  disp(alloc)
18  figure
19  % -------- Graph 1: RB Allocation per User --------
```

Figures

Figure 1 | Figure 2



Command Window

```
>> untitled8
Total Resource Blocks:
    55

RB Allocation per User:
    10     9     9     9     9     9
```

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Save As | Show Code

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

FILE

LABELS AND ANNOTATIONS

/ › MATLAB Drive

Files

Name

- untitled9.m
- untitled8.m
- untitled7.m
- untitled6.m
- untitled5.m
- untitled4.m
- untitled3.m
- untitled2.m
- untitled01.m
- untitled.m mfw.m
- untitled.m

Workspace

Name	Value	Size	Class
alloc	[12,10,9,8,11]	1×5	double
total_RB	50	1×1	double
users	[1,2,3,4,5]	1×5	double
util	[24,20,18,16,...	1×5	double

Editor

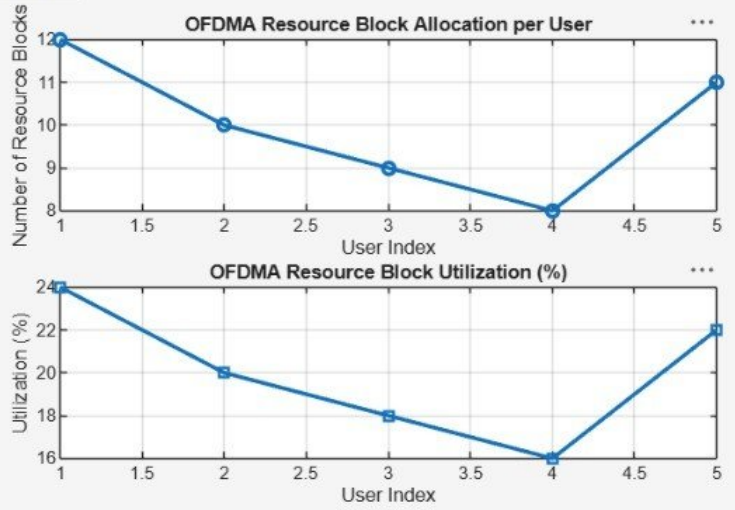
ed5.m | untitled2 | untitled6.m | untitled7.m | untitled8.m | untitled9.m

/MATLAB Drive/untitled9.m

```
1   clc; clear; close all;
2   % Total system resource blocks
3   total_RB = 50;
4   % Number of active users
5   users = 1:5;
6   % RB allocation to users (example scenario)
7   alloc = [12 10 9 8 11];
8   % Resource Block Utilization (%)
9   util = (alloc / total_RB) * 100;
10  disp('Resource Block Utilization (%) per User:')
11  disp(util')
12  figure
13  % -------- Graph 1: RB Allocation --------
14  subplot(2,1,1)
15  plot(users, alloc, '-o', 'LineWidth', 2)
16  title('OFDMA Resource Block Allocation per User')
17  xlabel('User Index')
18  ylabel('Number of Resource Blocks')
19  grid on
20  % -------- Graph 2: Utilization (%) --------
```

Figures

Figure 1



Command Window

```
    20
    18
    16
    22

>>
```

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Show Code

Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

FILE

LABELS AND ANNOTATIONS

/ › MATLAB Drive

Files

Name

- untitled10.m
- untitled9.m
- untitled8.m
- untitled7.m
- untitled6.m
- untitled5.m
- untitled4.m
- untitled3.m
- untitled2.m
- untitled01.m
- untitled.m mfw.m

Workspace

Name	Value	Size	Class
alloc_RB	[12,10,9,8,11]	1×5	double
mod_eff	[2,4,6,4,2]	1×5	double
rb_bw	180000	1×1	double
throughput	[4.3200,7.200...	1×5	double
users	[1,2,3,4,5]	1×5	double

Editor

d6.m | untitled7.m | untitled8.m | untitled9.m | untitled7 | untitled10.m

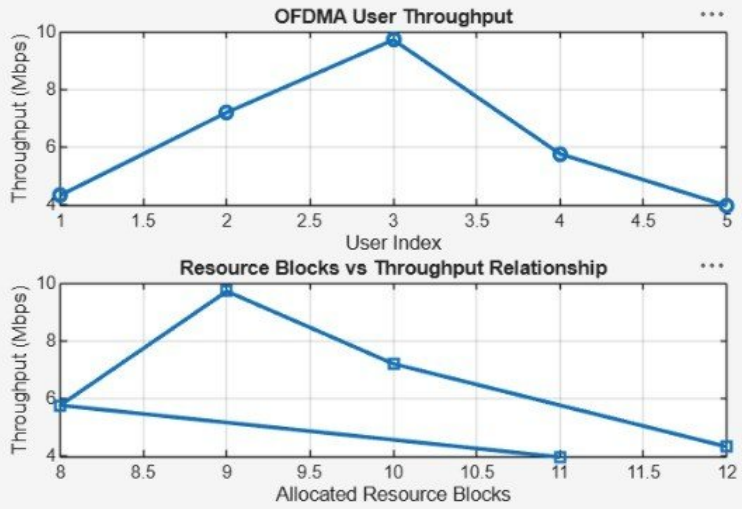
/MATLAB Drive/untitled10.m

```
clc; clear; close all;
% System parameters
rb_bw = 180e3; % Bandwidth per RB (180 kHz)
users = 1:5;
% Allocated Resource Blocks per user
alloc_RB = [12 10 9 8 11];
% Modulation efficiency (bits/symbol)
mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
% Throughput calculation (Mbps)
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')
disp(throughput')
figure
% -------- Graph 1: Throughput per User --------
subplot(2,1,1)

plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
```

Figures

Figure 1



Command Window

```
    7.2000
    9.7200
    5.7600
    3.9600

>>
```